BEAM DESIGN SUMMARY

Project Name	:	G+2 RB PLINTH BEAM DESIGN
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-1.981 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 03:00:45
Level Designed	:	1.981 m

Beam No		:	**B1**	
Group No		:	**G1**	
Analysis Reference(Member)	1.981 m	:	1	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	59.62	48.71	31.43	72.46	28.34	80.31
PtClc (%)	0.438	0.35	0.219	0.547	0.2	0.617
AstCalc (sqmm)	372.84	298.25	186.54	465.56	170.2	525.32
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	58.76	50.08	73.63
Asv Torsion (sqmm/m)	-	175.9	250.46
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B2**

Group No		:	**G1**	
Analysis Reference(Member)	1.981 m	:	2	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	37.16	35.32	46.55	70.79	38.05	94.97
PtClc (%)	0.262	0.248	0.334	0.533	0.268	0.757
AstCalc (sqmm)	222.82	211.05	283.86	453.19	228.48	644.27
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	66.1	44.47	68.03
Asv Torsion (sqmm/m)	228.92	145.91	220.47
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B3**	
Group No		:	**G1**	
Analysis Reference(Member)	1.981 m	:	194	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	20.85	20.85	20.85	55.63	46.06	29.59
PtClc (%)	0.2	0.2	0.2	0.406	0.33	0.206
AstCalc (sqmm)	170.2	170.2	170.2	345.15	280.65	175.08
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	115.11	111.68	108.25
Asv Torsion (sqmm/m)	744.09	733.23	722.36
Asv Reqd (sqmm/m)	437.37	411.69	386
Asv Prv (sqmm/m)	744.74	744.74	744.74
Reinforcement	2L-T8 @ 135	2L-T8 @ 135	2L-T8 @ 135

SFR : -

Beam No		**:**	**B4**	
Group No		**:**	**G2**	
Analysis Reference(Member)	1.981 m	:	21	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	69.72	47.44	44.7	71.25	34.35	71
PtClc (%)	0.523	0.34	0.319	0.537	0.241	0.534
AstCalc (sqmm)	445.36	289.76	271.63	456.58	204.92	454.74
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	46.75	42.99	54.77
Asv Torsion (sqmm/m)	154.02	142.81	180.09
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B5**	
Group No		:	**G2**	
Analysis Reference(Member)	1.981 m	:	22	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	51.52	31.81	50.84	59.08	51.25	102.01
PtClc (%)	0.373	0.222	0.367	0.434	0.371	0.829
AstCalc (sqmm)	317.14	188.92	312.55	369.07	315.3	705.52
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	43.73	45.58	57.22
Asv Torsion (sqmm/m)	145.04	149.48	185.02
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B6**

Group No		:	**G2**	
Analysis Reference(Member)	1.981 m	:	195	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.88	16.88	19.9	78.05	58.98	28.28
PtClc (%)	0.2	0.2	0.2	0.597	0.433	0.2
AstCalc (sqmm)	170.2	170.2	170.2	507.89	368.41	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	69.85	69.02	66.44
Asv Torsion (sqmm/m)	266.01	263.39	255.21
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B7**	
Group No		:	**G3**	
Analysis Reference(Member)	1.981 m	:	41	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	74.37	48.67	51.79	75.56	35.94	79.11
PtClc (%)	0.564	0.35	0.375	0.574	0.253	0.606
AstCalc (sqmm)	479.9	297.96	318.98	488.85	214.98	516
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	50.1	44.9	58.69
Asv Torsion (sqmm/m)	166.78	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B8**	
Group No		**:**	**G3**	
Analysis Reference(Member)	1.981 m	:	42	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	59.38	34.37	56.87	66.37	51.33	104.58
PtClc (%)	0.436	0.241	0.416	0.495	0.371	0.851
AstCalc (sqmm)	371.16	205.06	353.7	420.94	315.85	723.86
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	46.01	49.33	61.11
Asv Torsion (sqmm/m)	-	166.75	204.03
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B9**	
Group No		:	**G3**	
Analysis Reference(Member)	1.981 m	:	196	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	19.42	19.42	19.42	68.86	52.49	30.48
PtClc (%)	0.2	0.2	0.2	0.516	0.38	0.212
AstCalc (sqmm)	170.2	170.2	170.2	438.98	323.71	180.66
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	70.66	69.83	67.25
Asv Torsion (sqmm/m)	321.72	319.1	310.92
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No : **B10**

Group No		:	**G4**	
Analysis Reference(Member)	1.981 m	:	61	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	70.28	53.04	45.42	83.3	33.38	93.44
PtClc (%)	0.528	0.385	0.325	0.645	0.234	0.742
AstCalc (sqmm)	449.44	327.41	276.37	548.72	198.8	631.31
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	67.7	52.72	76.28
Asv Torsion (sqmm/m)	227.5	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B11**	
Group No		:	**G4**	
Analysis Reference(Member)	1.981 m	:	62	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	51.19	35.51	50.79	79.94	44.32	108.25
PtClc (%)	0.37	0.249	0.367	0.614	0.316	0.88
AstCalc (sqmm)	314.91	212.28	312.21	522.47	269.13	748.59
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	65.01	51.52	75.07
Asv Torsion (sqmm/m)	-	167.74	242.3
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B12**	
Group No		:	**G4**	
Analysis Reference(Member)	1.981 m	:	197	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	10.56	10.56	15.63	66.11	48.74	18.15
PtClc (%)	0.2	0.2	0.2	0.492	0.351	0.2
AstCalc (sqmm)	170.2	170.2	170.2	419.03	298.41	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	77.67	77.39	74.98
Asv Torsion (sqmm/m)	366.6	365.72	358.09
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B13**	
Group No		:	**G5**	
Analysis Reference(Member)	1.981 m	:	191	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	18.39	21.73	13.86	10.96	4.6	14.16
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	Mid	Right
Vut (kN)	38.56	22.95	42.55
Asv Torsion (sqmm/m)	194.12	134.05	206.47
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No : **B14**

Group No		:	**G5**	
Analysis Reference(Member)	1.981 m	:	193	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	8.38	14.42	11.38	12.98	11.04	27.69
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	30.02	21.28	35.23
Asv Torsion (sqmm/m)	146.17	119.14	153.21
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B15**	
Group No		:	**G5**	
Analysis Reference(Member)	1.981 m	:	198	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	16.68	16.68	16.68	36	31.68	26.27
PtClc (%)	0.2	0.2	0.2	0.253	0.221	0.2
AstCalc (sqmm)	170.2	170.2	170.2	215.37	188.16	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	226.2	226.2	226.2
Reinforcement	2-T12	2-T12	2-T12	2-T12	2-T12	2-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	79.67	76.24	72.81
Asv Torsion (sqmm/m)	505.08	494.46	483.84
Asv Reqd (sqmm/m)	357.75	332.06	306.38
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B16**	
Group No		:	**G6**	
Analysis Reference(Member)	1.981 m	:	189	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	14.23	14.23	14.23	25.16	53.17	70.75
PtClc (%)	0.2	0.2	0.2	0.2	0.386	0.532
AstCalc (sqmm)	170.2	170.2	170.2	170.2	328.3	452.89
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	74.89	81.66	85.63
Asv Torsion (sqmm/m)	451.36	472.54	484.97
Asv Reqd (sqmm/m)	254.81	254.81	282.82
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B17**	
Group No		:	**G6**	
Analysis Reference(Member)	1.981 m	:	105	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	25.03	25.97	26.86	82.99	24.21	68.3
PtClc (%)	0.2	0.2	0.2	0.642	0.2	0.511
AstCalc (sqmm)	170.2	170.2	170.2	546.29	170.2	434.94
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	Mid	Right
Vut (kN)	64.17	37.19	61.22
Asv Torsion (sqmm/m)	208.09	123.64	208.82
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B18**

Group No		:	**G6**	
Analysis Reference(Member)	1.981 m	:	93	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	21.21	26.62	25.16	77.79	16.05	70.53
PtClc (%)	0.2	0.2	0.2	0.594	0.2	0.53
AstCalc (sqmm)	170.2	170.2	170.2	505.84	170.2	451.32
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	62.3	31.88	59.25
Asv Torsion (sqmm/m)	199.67	104.48	189.7
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B19**	
Group No		:	**G6**	
Analysis Reference(Member)	1.981 m	:	81	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	19.09	33.07	30.28	83.8	18.02	68.73
PtClc (%)	0.2	0.231	0.211	0.649	0.2	0.515
AstCalc (sqmm)	170.2	196.87	179.4	552.66	170.2	438.09
Ast Prv (sqmm)	226.2	226.2	226.2	603.18	603.18	603.18
Reinforcement	2-T12	2-T12	2-T12	3-T16	3-T16	3-T16

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	68.24	38.28	58.57
Asv Torsion (sqmm/m)	229.41	135.65	187.7
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B20**	
Group No		:	**G7**	
Analysis Reference(Member)	1.981 m	:	190	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	12.06	9.57	9.57	22.54	69.72	97.75
PtClc (%)	0.2	0.2	0.2	0.2	0.523	0.785
AstCalc (sqmm)	170.2	170.2	170.2	170.2	445.32	668.04
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	44.73	50.97	52.97
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B21**	
Group No		:	**G7**	
Analysis Reference(Member)	1.981 m	:	106	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	Mid	Right	Left	Mid	Right
Mud (kNm)	41.76	30.7	48.61	83.07	32.59	62.99
PtClc (%)	0.297	0.214	0.35	0.643	0.228	0.466
AstCalc (sqmm)	252.43	181.98	297.55	546.96	193.82	396.63
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	55.96	38.67	47.18
Asv Torsion (sqmm/m)	180.93	126.19	153.44
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No	:	**B22**
Group No	:	**G7**

Analysis Reference(Member)	1.981 m	:	94	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	30.48	25.56	33.97	72.4	22.85	59.14
PtClc (%)	0.212	0.2	0.238	0.547	0.2	0.434
AstCalc (sqmm)	180.62	170.2	202.52	465.13	170.2	369.54
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	49.01	29.77	44.41
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B23**	
Group No		**:**	**G7**	
Analysis Reference(Member)	1.981 m	:	82	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	28.94	32.06	41.72	73.24	21.98	57.62
PtClc (%)	0.201	0.224	0.296	0.554	0.2	0.422
AstCalc (sqmm)	171.04	190.53	252.15	471.44	170.2	358.93
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	52.04	32.98	43.13
Asv Torsion (sqmm/m)	170.54	110.2	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B24**	
Group No		:	**G8**	
Analysis Reference(Member)	1.981 m	:	192	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	21.71	12.43	12.43	22	64.76	88.83
PtClc (%)	0.2	0.2	0.2	0.2	0.481	0.697
AstCalc (sqmm)	170.2	170.2	170.2	170.2	409.29	593.15
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	87.19	92.04	94.05

Asv Torsion (sqmm/m)	494.85	510.17	516.55
Asv Reqd (sqmm/m)	254.81	264.62	279.7
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B25**	
Group No		:	**G8**	
Analysis Reference(Member)	1.981 m	:	107	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	34.54	32.07	38.82	96.65	31.34	79.87
PtClc (%)	0.242	0.224	0.274	0.774	0.219	0.613
AstCalc (sqmm)	206.14	190.56	233.42	658.56	186.04	521.87
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	70.37	43.38	66.83
Asv Torsion (sqmm/m)	232.16	146.74	232.8
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B26**
Group No		:	**G8**
Analysis Reference(Member)	1.981 m	:	95

Breadth	:	230	mm
Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	25.85	31.73	30.1	90.01	22.16	80.65
PtClc (%)	0.2	0.221	0.209	0.708	0.2	0.62
AstCalc (sqmm)	170.2	188.43	178.26	602.85	170.2	527.95
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	69.53	39.05	62.35
Asv Torsion (sqmm/m)	235.33	139.81	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B27**	
Group No		**:**	**G8**	
Analysis Reference(Member)	1.981 m	:	83	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	27.8	40.79	40.1	96.53	23.95	82.21
PtClc (%)	0.2	0.289	0.284	0.773	0.2	0.635
AstCalc (sqmm)	170.2	246.13	241.68	657.54	170.2	540.17
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	76.42	46.46	69.84
Asv Torsion (sqmm/m)	272.47	177.64	254.93
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B28**	
Group No		:	**G9**	
Analysis Reference(Member)	1.981 m	:	199	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	14.01	14.01	14.01	23.89	37.22	47.89
PtClc (%)	0.2	0.2	0.2	0.2	0.262	0.344
AstCalc (sqmm)	170.2	170.2	170.2	170.2	223.2	292.74
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	63	69.61	74.45

Asv Torsion (sqmm/m)	405.1	425.56	434.73
Asv Reqd (sqmm/m)	254.81	254.81	273.69
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B29**	
Group No		:	**G9**	
Analysis Reference(Member)	1.981 m	:	200	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	9.09	9.81	3.16	39.1	16.02	19.54
PtClc (%)	0.2	0.2	0.2	0.276	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	235.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	32.28	14.07	28.97
Asv Torsion (sqmm/m)	122.75	46.94	128.02
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B30**
Group No		:	**G9**
Analysis Reference(Member)	1.981 m	:	201

Breadth	:	230	mm
Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	6.75	14.55	6.71	20.26	1.57	23.07
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	30.31	14.59	31.42
Asv Torsion (sqmm/m)	113.89	65.24	117.32
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B31**
Group No		**:**	**G9**
Analysis Reference(Member)	1.981 m	:	202
Breadth		:	230 mm
Depth		:	430 mm
Concrete Grade		:	M25 N/sqmm
Grade Of Steel (Main)		:	Fe550 N/sqmm
Grade Of Steel (Shear)		:	Fe550 N/sqmm
Top/Bottom Clear Cover		:	30 mm
Side Clear Cover		:	30 mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016
Beam Type		:	Regular Beam

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	10.34	23.91	20.32	27.77	4.47	15.42
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	47.55	30.99	40.04
Asv Torsion (sqmm/m)	219.82	168.57	196.59
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -